

\$sampleRate (aggregation)

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Definition

`$sampleRate`

New in version 4.4.2.

Matches a random selection of input documents. The number of documents selected approximates the sample rate expressed as a percentage of the total number of documents.

The `$sampleRate` operator has the following syntax:

```
{ $sampleRate: <non-negative float> }
```

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Behavior

The selection process uses a uniform random distribution. The sample rate is a floating point number between 0 and 1, inclusive, which represents the probability that a given document will be selected as it passes through the pipeline.

For example, a sample rate of `0.33` selects roughly one document in three.

This expression:

```
{ $match: { $sampleRate: 0.33 } }
```



is equivalent to using the `$rand` operator as follows:

```
{ $match: { $expr: { $lt: [ { $rand: {} }, 0.33 ] } } }
```



Repeated runs on the same data will produce different outcomes since the selection process is non-deterministic. In general, smaller datasets will show more variability in the number of documents selected on each run. As collection size increases, the number of documents chosen will approach the expected value for a uniform random distribution.

NOTE

If an exact number of documents is required from each run, the `$sample`

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operator should be used instead of `$sampleRate`.

Examples

This code creates a small collection with 100 documents.

```
N = 100
bulk = db.collection.initializeUnorderedBulkOp()
for ( i = 0; i < N; i++) { bulk.insert( { _id: i, r: 0 } ) }
bulk.execute()
```



The `$sampleRate` operator can be used in a pipeline to select random documents from the collection. In this example we use `$sampleRate` to select about one third of the documents.

```
db.collection.aggregate(
  [
    { $match: { $sampleRate: 0.33 } },
    { $count: "numMatches" }
  ]
)
```



This is the output from 5 runs on the sample collection:

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```
{ "numMatches" : 38 }  
{ "numMatches" : 36 }  
{ "numMatches" : 29 }  
{ "numMatches" : 29 }  
{ "numMatches" : 28 }
```



TIP

See also:

- `$sample`
- `$rand`

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